

ALERT CS, NU, Canada

WMO: 713550

Lat: 82.493N

Lon: 62.350W

Elev: 215

StdP: 14.58

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-40.2	-36.7	-47.9	0.3	-40.1	-44.1	0.4	-36.7	42.4	-4.4	36.4	-2.8	3.2	280	1.139

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	58.4	48.2	53.2	45.7	48.2	42.8	48.3	57.2	45.9	53.5	42.8	48.1	19.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
41.1	38.3	52.9	39.0	35.3	48.8	37.3	32.9	45.6	19.4	57.5	18.1	53.3	16.6	48.3	54.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.2	29.4	24.5	DB	-43.7	62.4	4.1	4.3	-46.6	65.5	-49.1	68.0	-51.4	70.4	-54.4	73.6	(4)
(5)			WB	-43.8	51.7	4.1	2.1	-46.8	53.3	-49.2	54.5	-51.5	55.7	-54.5	57.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.0	-19.0	-20.6	-21.5	-8.9	12.7	32.7	39.6	35.1	18.9	2.8	-9.5	-15.9		
	DBStd	23.88	9.27	10.19	10.23	10.07	8.95	7.01	6.56	6.54	7.78	10.05	8.60	9.95		
	HDD50	16795	2140	1978	2217	1766	1155	520	332	464	933	1462	1784	2044		
	HDD65	22256	2605	2398	2682	2216	1620	969	787	926	1383	1927	2234	2509		
	CDD50	14	0	0	0	0	0	1	10	3	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	6.7	7.1	6.0	5.5	5.2	6.0	7.4	8.1	7.5	6.8	7.6	7.2	6.0		
	PrecAvg	6.8	0.3	0.2	0.3	0.4	0.5	0.5	1.1	1.0	1.1	0.6	0.4	0.3		
	PrecMax	11.5	1.3	0.9	1.1	1.5	1.1	1.9	2.2	2.3	2.3	1.7	1.0	0.9		
(15) Precipitation	PrecMin	3.7	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.2	0.0	0.1		
	PrecStd	1.4	0.2	0.2	0.2	0.4	0.3	0.4	0.6	0.6	0.6	0.4	0.3	0.2		
	DB	15.4	17.5	14.2	19.9	36.4	57.2	64.3	59.3	37.5	29.4	21.0	24.7			
	MCWB	13.3	15.1	12.3	18.0	33.0	48.0	50.4	48.9	33.6	26.5	17.8	22.9			
	2% DB	4.2	5.0	4.4	14.6	30.8	50.3	61.0	54.7	34.7	22.8	12.7	13.4			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	2.8	4.0	3.2	13.0	27.7	44.2	48.9	46.5	31.6	20.7	10.9	12.1			
	5% DB	-0.6	-1.9	-2.1	10.1	27.9	45.6	56.7	50.0	31.8	20.0	7.3	4.0			
	MCWB	-1.4	-2.7	-2.8	8.9	25.9	40.9	47.4	43.3	29.5	18.4	6.1	3.1			
	10% DB	-5.2	-6.6	-6.6	5.9	25.6	42.8	51.2	45.8	29.3	16.8	3.3	-2.2			
	MCWB	-5.8	-7.1	-7.1	4.9	24.1	39.3	45.0	40.9	27.8	15.5	2.3	-2.9			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	13.3	15.1	12.5	18.4	33.0	48.4	52.0	49.8	34.0	26.5	18.0	23.1			
	MCDB	15.5	17.4	14.3	19.8	36.3	56.6	63.0	58.3	36.8	29.0	21.0	24.7			
	2% WB	2.8	4.0	3.1	13.2	28.0	44.2	49.4	47.0	31.6	21.1	11.0	12.2			
	MCDB	3.9	4.9	4.4	14.6	30.0	50.0	59.4	54.8	33.9	22.6	12.5	13.7			
	5% WB	-1.4	-2.7	-2.8	8.9	26.1	41.2	47.3	43.4	29.7	18.5	6.3	3.0			
(30) Mean Daily Temperature Range	MCDB	-0.7	-1.9	-2.3	10.0	27.8	45.3	56.4	49.7	31.4	20.0	7.2	3.7			
	10% WB	-5.8	-7.2	-7.1	4.9	24.2	39.1	44.8	41.0	27.9	15.5	2.5	-2.9			
	MCDB	-5.2	-6.8	-6.6	5.8	25.7	42.8	51.4	45.7	29.3	16.8	3.3	-2.2			
	MDBR	10.9	11.2	10.8	10.4	7.5	7.0	9.2	8.3	7.9	9.9	10.5	10.8			
	5% DB	16.8	18.9	16.0	14.5	9.2	12.4	18.6	15.8	10.1	12.1	14.7	16.9			
(36) Clear-Sky Solar Irradiance	MCWBR	16.0	18.1	15.2	13.6	7.4	8.3	11.1	10.3	8.0	10.7	13.6	15.8			
	MCDBR	16.5	18.4	15.9	14.2	8.9	12.3	18.0	15.5	9.4	12.0	14.3	16.9			
	MCWBR	15.7	17.6	15.2	13.3	7.2	8.5	11.3	10.4	7.5	10.6	13.4	15.8			
	taub	N/A	N/A	0.173	0.250	0.273	0.299	0.289	0.268	0.193	N/A	N/A	N/A			
	taud	N/A	N/A	1.975	2.125	2.176	2.231	2.397	2.408	2.070	N/A	N/A	N/A			
(42) All-Sky Solar Radiation	Ebn at Noon	0	0	161	235	259	257	256	236	156	0	0	0			
	Edh at Noon	0	0	10	23	29	30	23	18	10	0	0	0			
(44) All-Sky Solar Radiation	RadAvg	0	0	164	1083	1993	2272	1910	1059	303	0	0	0			
	RadStd	0	0	15	60	112	108	135	108	27	0	0	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page